

WTW 158 Worksheet on **Curve sketching**

1. Do Exercise 4.3 p300 Numbers 25, 26, 27, 28

2. Sketch the following curves by systematically working through the following aspects:

D - Domain

I - Intercepts with axes

S - Symmetry - Is the function even/odd/neither

A - Asymptotes - Vertical/horizontal

D - Where the function decreases/ increases

E - Extremes

C - Concavity

I - Inflection points

i. Exercise 4.5 p321 no 7 and 13.

ii. $f(x) = x - 2 \sin x$ for $0 \leq x \leq 2\pi$

iii. $f(x) = \cos^2 x + \cos x$ for $0 \leq x \leq 2\pi$

iv. $f(x) = \frac{x}{\sqrt{x^2+2}}$ given that $f'(x) = \frac{2}{(x^2+2)^{\frac{3}{2}}}$ and $f''(x) = -\frac{6x}{(x^2+2)^{\frac{5}{2}}}$

v. $f(x) = \frac{x^2-9}{x^2-4}$ given that $f'(x) = \frac{10x}{(x^2-4)^2}$ and $f''(x) = \frac{-10(3x^2+4)}{(x^2-4)^3}$